

Periodic BC $\Rightarrow$ cyclic tridiagonal $\Rightarrow$ Cyclic Thomas

b	c			a_1
a	b	c		
	a	b	c	
		a	b	c
c_n			a	b

$$A\mathbf{x} = \mathbf{d}$$

$$A = \begin{bmatrix} b_1 & c_1 & & & a_1 \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & \ddots & \\ & & \ddots & \ddots & c_{n-1} \\ c_n & & & a_n & b_n \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{bmatrix}$$

a_1 and c_n are generally nonzero, so Thomas cannot be applied to A directly.

Cyclic Thomas (Sherman–Morrison). In order to solve $A\mathbf{x} = \mathbf{d}$, Choose $\gamma \in \mathbb{R}$ (e.g. $\gamma = -b_1$) and write $A = B + \mathbf{u}\mathbf{v}^\top$ with B purely tridiagonal:

$$B = \begin{bmatrix} b_1 - \gamma & c_1 & & & 0 \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & \ddots & \\ & & \ddots & \ddots & c_{n-1} \\ 0 & & & a_n & b_n - c_n a_1 / \gamma \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} \gamma \\ 0 \\ 0 \\ \vdots \\ c_n \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ a_1 / \gamma \end{bmatrix}.$$

The solution is then given by

$$B\mathbf{y} = \mathbf{d}, \quad B\mathbf{q} = \mathbf{u}, \quad \mathbf{x} = \mathbf{y} - \frac{\mathbf{q}(\mathbf{v}^\top \mathbf{y})}{1 + \mathbf{v}^\top \mathbf{q}}.$$

PEC BC $\Rightarrow$ Dirichlet row in the line solve

Perfect electric conductor: $\hat{\mathbf{n}} \times \mathbf{E} = 0$, $\hat{\mathbf{n}} \cdot \mathbf{B} = 0$.

If the PEC boundary is normal to x : $E_y = 0$, $E_z = 0$.

$$\begin{bmatrix} b_0 & c_0 & 0 & \cdots & 0 & 0 & \cdots & 0 & \alpha \\ a_1 & b_1 & c_1 & \cdots & & & & & 0 \\ & 0 & a_2 & b_2 & \ddots & & & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & & & & \vdots \\ & \cdots & a_{m-1} & b_{m-1} & c_{m-1} & 0 & \cdots & & \\ & \cdots & 0 & 0 & \boxed{1} & 0 & \cdots & & \\ & \cdots & 0 & 0 & a_{m+1} & b_{m+1} & c_{m+1} & \cdots & \\ 0 & \cdots & & & a_{N-2} & b_{N-2} & c_{N-2} & 0 & \\ \beta & 0 & \cdots & & 0 & 0 & a_{N-1} & b_{N-1} \end{bmatrix} \begin{bmatrix} E_{y,0} \\ E_{y,1} \\ \vdots \\ E_{y,m-1} \\ \boxed{E_{y,m}} \\ E_{y,m+1} \\ \vdots \\ E_{y,N-1} \end{bmatrix} = \begin{bmatrix} r_0 \\ r_1 \\ \vdots \\ r_{m-1} \\ \boxed{0} \\ r_{m+1} \\ \vdots \\ r_{N-1} \end{bmatrix}$$

Row m is replaced by the identity: $E_{y,m} = 0$ (and likewise $E_{z,m} = 0$). Off-diagonals in that row are dropped, and the RHS is 0. Domain walls normal to x are the endpoint cases $m = 0$ and $m = N - 1$; the remaining interior rows stay tridiagonal and are solved by ordinary Thomas.